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Research article

The as-cast $Al_xCrTaTi$ refractory medium entropy alloys with good room-temperature mechanical properties and high-temperature oxidation resistance

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ABSTRACT

Refractory medium entropy alloys (RMEAs) usually show good mechanical properties at high temperature. Unfortunately, their very poor room-temperature plasticity and high-temperature oxidation resistance still greatly limit their practical application. In this work, the room-temperature mechanical properties and high-temperature oxidation resistance of $Al_xCrTaTi$ RMEAs (x = 0, 0.15, 0.25, 0.35) are systematically studied. It is found that the $Al_xCrTaTi$ RMEAs exhibit very high yield strength of about 1700 MPa and plasticity of above 10%. For Al-bearing RMEAs, a composite oxide layer containing Al_2O_3 is formed on the alloy surface. Under the synergistic effect with the $CrTaO_4$ oxide layer in the middle layer, $Al_{0.25}CrTaTi$ alloy possesses good oxidation resistance at 1000 °C for 100 h with the oxidation weight gain of only 7.3 mg/cm^2.
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1. Introduction

Compositionally complex alloys (CCAs) are a new type of metals found near the center of the phase diagrams, which are completely different from the conventional alloys that tend to cluster around the corners or edges of the phase diagrams. The discovery of CCAs broken the design concept of the traditional alloys and provided a new strategy for the development of materials [1–3]. Among the CCAs systems, it was found that refractory high entropy alloys (RHEAs) were obtained by combining a variety of refractory elements at nearly equal atomic ratios, which exhibited remarkable mechanical properties at elevated temperatures, and even their service temperature exceeded the existing nickel-based superalloys [4–6]. The RHEAs have become one of the superalloys with great potential as to have attracted extensive attention from the field of science and industry worldwide [7].

However, refractory high-entropy alloys usually exhibit poor room temperature plasticity (< 10%) and relatively high densities, such as a typical RHEAs of $V_{20}Nb_{20}Mo_{20}Ta_{20}W_{20}$ with room temperature plasticity less than 2% and a density as high as 12.36 g/cm^3 [8]. From the aspect of deformation mechanism, the screw dislocation that dominates the plastic deformation is uneasy to motivate due to the high friction stress in BCC RMEA [9]. From the electronic structure perspective, the high number of valence electrons of RMEA shifts the Fermi level lower the band structure, resulting the less strain is required and the shear instability arises earlier [10]. Therefore, the RHEAs usually exhibit very poor room-temperature plasticity. In addition, the refractory metals are normally difficult to form a continuous dense oxide film at the elevated temperature, which results the severe oxidation of RHEAs during the high-temperature service [11]. Therefore, it is an important and urgent issue to develop the refractory CCAs with good mechanical properties. In recent work, we found a new refractory medium-entropy alloy (RMEA) of CrTaTi with excellent room temperature mechanical properties. The alloy density was reduced to 9.89 g/cm^3, and also has a high compression yield strength (1498 MPa) and room temperature plasticity (> 20%), respectively. However, it still exhibits a relatively poor long-term high-temperature oxidation resistance, which is a crucial aspect for practical applications of RMEAs [12–14], such as aerospace, nuclear energy, petroleum industry fields.

Previously studies have shown that Al was an ideal antioxidant element, and α-Al_2O_3 in its oxides was a dense and stable oxide, which could effectively prevent the inward diffusion of oxygen at high temperatures and improve the oxidation resistance of alloys [15–17]. In addition, Al is a light element, and the introduction of low-density Al could effectively reduce the density of the alloy and improve the specific strength of the materials [18]. At the same time,

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it can also be used as a strong BCC stabilizing element in refractory high-entropy alloys [19]. In this work, the microstructure, mechanical properties and high temperature oxidation behavior of $Al_xCrTaTi$ alloys with different Al contents were systematically studied. It is expected that the density of the alloy can be reduced by introducing low-density Al without losing the excellent comprehensive mechanical properties at room temperature. At the same time, it can effectively improve the high-temperature oxidation resistance of the alloy, which provides theoretical basis and technical supports for expanding the application space of refractory high-entropy alloys.

2. Experimental methods

AlxCrTaTi (x = 0, 0.15, 0.25 and 0.35, atomic molar ratio) refractory medium-entropy alloy ingots with different Al contents were prepared through vacuum arc-melting technology by using the high-purity metals (≥99.9 wt%) of Al, Cr, Ta, Ti in a high-purity argon atmosphere. The alloy rod with 3 mm in diameter and 60 mm in length was fabricated by copper mold suction casting. For room-temperature mechanical compression, the sample rod with a diameter of 3 mm and 6 mm in length was used by sandpaper with different meshes to smooth the end surface. The test was carried out with a CMT5504 universal mechanical testing machine, and the compressive strain rate was 10^{-3} s^{-1}. In order to ensure the reliability of the data, the compression experiment of each group of components was repeated at least 5 times. The Vickers hardness of the alloy was measured by a HVS-1000 A microhardness tester, with a pressure of 1.96 N and a holding pressure of 15 s, and at least 9 points were measured for each sample. The density of the rod samples was measured by the Archimedes drainage method. The samples used for oxidation experiment with a size of 3 mm × 3 mm× 2 mm were processed from the alloy ingots by wire cutting, then all the samples were polished with SiC sandpaper and cleaned with deionized water and ethanol thoroughly. Afterwards, the isothermal oxidation experiment was carried out at 1000 °C and the samples were set in the center of the muffle furnace under the experimental environment to be oxidized continuously for 10 h, 25 h, 50 h, and 100 h, respectively. And then all the samples were cooled down to room temperature within the furnace. The microstructure and phase composition of the original alloys and the oxide layers were investigated by X-ray diffraction (XRD, Shimadzu XRD-6100) and scanning electron microscopy (SEM, JSM-6610) with energy dispersive X-ray spectroscopy (EDS, Oxford Instruments), respectively. White light interferometer (Bruker, ContourGT-K) was used to analyze the surface roughness of the oxide layer of $Al_{0.15}CrTaTi$ ($Al_{0.15}$), $Al_{0.25}CrTaTi$ ($Al_{0.25}$), and $Al_{0.35}CrTaTi$ ($Al_{0.35}$) alloys after 100 h oxidation.

3. Results

3.1. Microstructure evolution

Fig. 1 shows the XRD patterns of as-cast Al_0, $Al_{0.15}$, $Al_{0.25}$ and $Al_{0.35}$ alloys, which are all mainly composed of BCC solid solution phase contained with Cr, Ta and Ti, and the Cr_2Ta Laves phase (JCPDS 05–0671). Although the phase composition of the alloy does not change significantly with the increase of Al content, it can be found that the peak position of BCC phase gradually shifts to the left with the increase of Al content. From the perspective of atomic radius, the atomic radii of Cr, Ta and Ti are 0.1249 nm, 0.1430 nm and 0.1462 nm respectively, while Al has a larger atomic radius (0.1432 nm) than the average radius (0.1380 nm) of the initial alloy, and multiple components will also lead to more serious lattice distortion of the alloy [20]. In addition, when the Al content is not very high, it can be dissolved in BCC solid solution to a certain extent. Due to the relatively large atomic radius of Al, the lattice distortion of BCC solid solution phase inevitably leads to the increase of lattice constant,

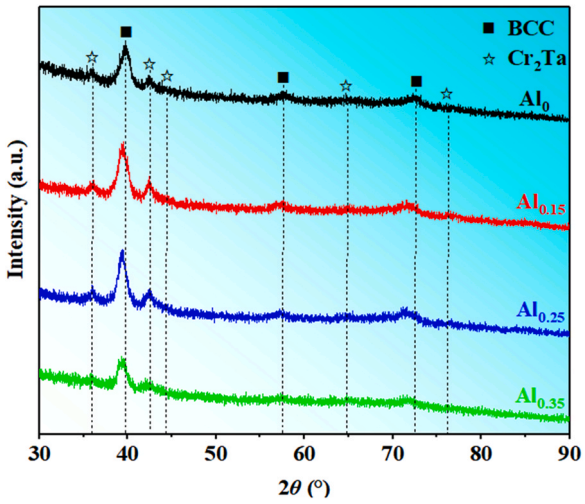


Fig. 1. XRD patterns of as-cast $Al_xCrTaTi$ RMEAs.

and finally, the peak position of the XRD pattern moves to the left with the increase of Al content. Although it appears that the peak intensity of Cr_2Ta decreases with the increase of Al, the Cr_2Ta phase content actually increases because this contradiction is caused by the decrease of the overall peak intensity of the sample. In order to further explore the phase distribution, Fig. 2 shows SEM images of the microstructure of $Al_xCrTaTi$ RMEAs. From the low-magnification backscattered electron images in Fig. 2(a-d), it can be seen that all the as-cast $Al_xCrTaTi$ alloys show obvious dendritic structures. Combined with energy spectrum analysis, it is notable that the matrix solid solution is bright dendrite with Ta-riched region, instead of Cr_2Ta which has a higher melting point than Ti and Cr. The darker interdendritic region (ID) is enriched with relatively low melting point elements of Cr, Ti, and Al. Furthermore, from the high magnification images of the microstructures shown in the insets of the Fig. 2(a-d), it can be clearly observed that the second phase is distributed in the interdendritic region. To obtain the more details of the microstructures in the ID region, the ID region of $Al_{0.35}$ is enlarged shown in Fig. 2e. It can be clearly seen that the smaller dendrites appear when the inter-dendritic region is enlarged. Furthermore, from the EDX elemental maps, the similar tendency of the distribution between Cr and Ta emerged in ID region, and they rich in the bright and smaller dendrites. Meanwhile, the distributions of Ti and Al also exhibit the same variation. However, there is different that the Ti and Al rich in the darker area in the ID region. Combined with XRD and energy spectrum analysis, it can be reasonably concluded that the Cr_2Ta phase is mainly distributed in the ID region. Therefore, during the non-equilibrium solidification, the dendrites riched in high melting point Ta are formed firstly, and then the interdendritic melts riched in Cr, Ti and Al are solidified with the precipitation of high melting point of Cr_2Ta dendrites. Within the limited resolution of XRD, there is no any of Cr_2Ti formation even whose melting temperature much lower than Cr_2Ta. This is because the mixing enthalpy of Cr between other refractory elements is more negative. The presence of Cr will reduce the enthalpy of the alloy system and the stability of the BCC phase, resulting in the appearance of the Laves phase in addition to the BCC matrix phase in the alloy. Although the addition of Al does not change the phase composition of the alloy, the shape of the Cr_2Ta phase changes significantly with different Al content. The Cr_2Ta phase in Al_0 alloy is rice-like and distributed uniformly in the ID region, and the Cr_2Ta phase in $Al_{0.15}$ alloy gradually becomes larger and elongated. With the continuous increase of Al, the elongated Cr_2Ta phase in $Al_{0.25}$ alloy becomes larger and tend to be connected together. Most of the Cr_2Ta phases in $Al_{0.35}$ alloy have been connected together, and the ID

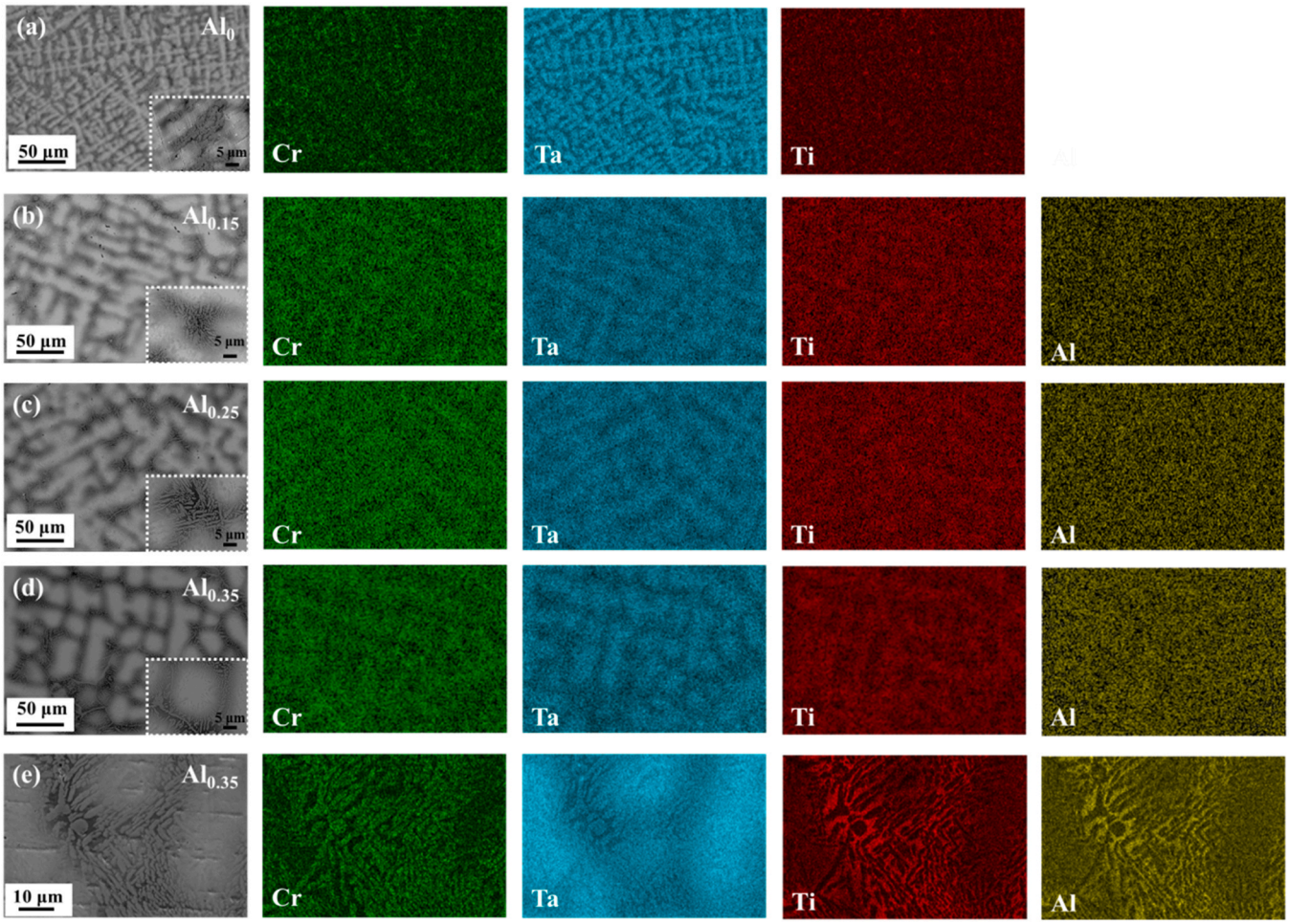


Fig. 2. Backscattered SEM images of the microstructure of as-cast $Al_xCrTaTi$ RMEAs.

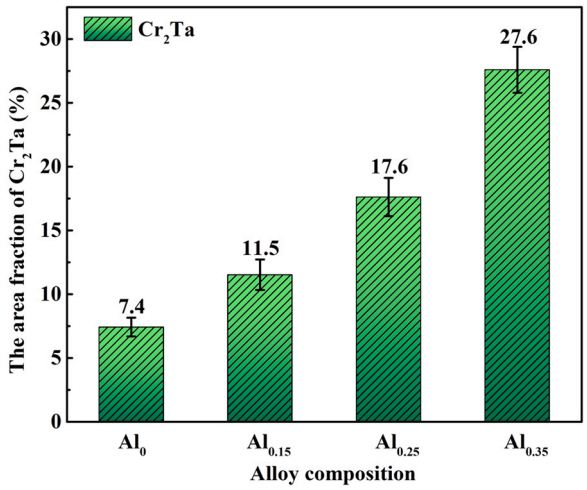


Fig. 3. The area fraction of Cr_2Ta phase in $Al_xCrTaTi$ RMEAs.

regions are cobweb-like distribution, and there are also agglomerated Cr_2Ta phases aggregated together. In addition, it can be clearly observed that the ID region increases with the increase of Al, while the Cr_2Ta phase is mainly distributed in the ID region, implying that the Cr_2Ta phase has a wider distribution region. Meanwhile, the area fraction of Cr_2Ta phase in several alloys is carefully counted by Image Tool. As shown in Fig. 3, with the increase of Al, the area fraction of the Cr_2Ta phase gradually increased from 7.42% of Al_0 to 27.59% of $Al_{0.35}$ alloys, indicating that the addition of Al promotes the formation of Cr_2Ta phase. This can be probably due to the production of more serious lattice distortion by addition of Al, which opens the channel for atomic diffusion [21]. Meanwhile, the lattice distortion can also reduce the diffusion activation energy, which is more conducive to the atomic diffusion [22]. Cr with the smallest atomic radius in the current alloy system is most prone to diffusion, and the mixing enthalpy of Cr and Ta is the lowest. Therefore, it is easier to generate Cr_2Ta phase. In addition, Al can also improve the stability of Cr_2Ta phase, so the addition of Al promotes the formation of Cr_2Ta phase [23,24].

3.2. Mechanical properties

Fig. 4a shows the microhardness of $Al_xCrTaTi$ RMEAs as a function of Al content. The microhardness of $Al_xCrTaTi$ alloys gradually increases with the increase of Al. Firstly, the addition of Al increases the lattice distortion of the BCC solid solution phase of the alloy matrix. The solid solution strengthening effect is significant, and the alloy hardness is improved obviously. Secondly, the increase of Al promotes the formation of much more Cr_2Ta Laves phase, and its area fraction increases significantly, thus the second-phase strengthening by Cr_2Ta Laves phase makes the alloys with much higher hardness. Furthermore, the relationship between the area fraction of Cr_2Ta phase (x) and the microhardness (y_{HV}) in present alloys is linearly fitted as shown in Fig. 4b. The linear relationship can be obtained as $y_{HV} = 650 + 1.40x$. The value of linear correlation coefficient R^2 is 0.961, indicating that there is a good relationship

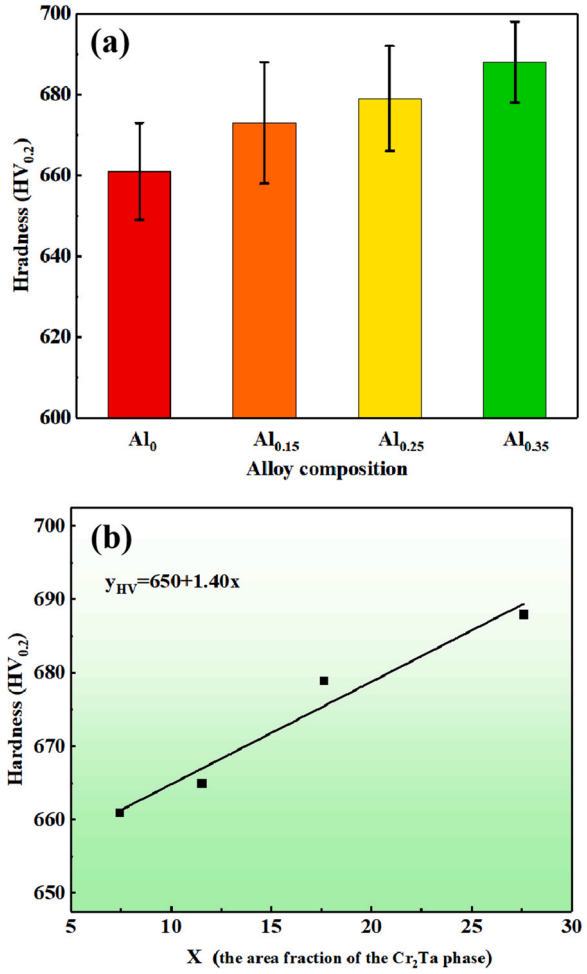


Fig. 4. Microhardness of $Al_xCrTaTi$ RMEAs (a); the relationship between microhardness and the content of Cr_2Ta phase in the alloy (b).

between the microhardness of $Al_xCrTaTi$ alloys and the content of Cr_2Ta phase.

Fig. 5a exhibits the engineering stress-strain curve of as-cast $Al_xCrTaTi$ RMEAs at room temperature. It can be seen that the addition of Al significantly improves the yield strength of the alloys, which are about 1498 MPa, 1541 MPa, 1669 MPa and 1780 MPa for Al_0, $Al_{0.15}$, $Al_{0.25}$ and $Al_{0.35}$ alloys, respectively. With the increase of Al, the fracture strain of the $Al_xCrTaTi$ RMEAs gradually decrease from 22% to 10%. The addition of Al increases the degree of lattice distortion of the alloy, which increases the resistance of dislocation movement. At the same time, Al also promotes the precipitation of more Cr_2Ta Laves phase in the alloy. Therefore, the synergistic effect of solution strengthening and precipitation strengthening of the alloy increases the strength of $Al_xCrTaTi$ alloys. Fig. 5b shows the room-temperature mechanical properties of the present RMEAs and some typical as-cast refractory CCA previously reported [25–35]. It can be seen that $Al_xCrTaTi$ alloys have a good combination of room-temperature strength and ductility. Although Al promotes the increase of the area fraction of hard and brittle Cr_2Ta Laves phase with usually resulting the poor plasticity of alloys, the fracture strain of present RMEAs still reaches above 10%. It should be noticed that the introduction of low-density Al effectively reduces the density of the alloy from 9.894 g/cm^3 of the original alloy Al_0 to 9.151 g/cm^3 of $Al_{0.35}$, resulting a very high specific strength of $Al_xCrTaTi$ refractory high entropy alloys as shown in Table 1.

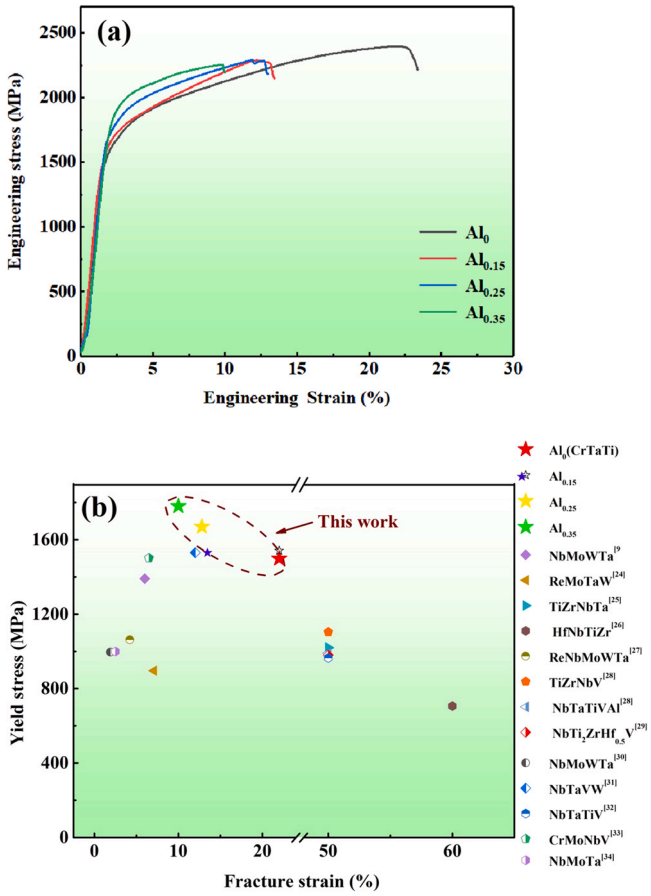


Fig. 5. The compressive stress-strain curves of the as-cast $Al_xCrTaTi$ RMEAs (a), the mechanical properties of present $Al_xCrTaTi$ alloys and some typical refractory CCAs reported previously [24–34] (b).

Table 1
Mechanical properties of as-cast $Al_xCrTaTi$ RMEAs at room temperature.

Alloys	$\sigma_{0.2}$ (MPa)	σ_c (MPa)	ε_f (%)	Vickers Hardness ($HV_{0.2}$)	ρ (g/cm^3)	specific strength (kN·m/kg)
Al_0	1498	2400	22	661±8	9.984	151.4
$Al_{0.15}$	1541	2289	13.4	665±15	9.384	164.2
$Al_{0.25}$	1669	2293	12.8	679±13	9.232	180.8
$Al_{0.35}$	1780	2256	10	688±9	9.151	194.5

$\sigma_{0.2}$: yield strength, σ_c: ultimate compressive strength, ε_f: fracture strain

3.3. Oxidation behavior

Fig. 6 shows the appearance of samples (a) and mass changes (b) of $Al_xCrTaTi$ RMEAs at 1000 ℃ for isothermal oxidation with different times. With the increase of oxidation time, the oxide layer on the surface of the alloy is peeled off under the action of external force held by tweezers after 25 h of oxidation of Al_0 sample. The thickness of the oxide layer increases significantly during 50 h of oxidation, which leads to the cracking at the corners of the sample due to the stress concentration. The macroscopic shape of the Al_0 sample becomes basically irregular, indicating that the alloy has been severely oxidized. The Al-bearing samples exhibit much better appearance than the original Al-free alloy in each oxidation stage. After the isothermal oxidation at 1000 °C for 25 h, the surfaces of the three alloys with different Al contents are smooth and flat, and the oxide layer shows good adhesion characteristics with complete on

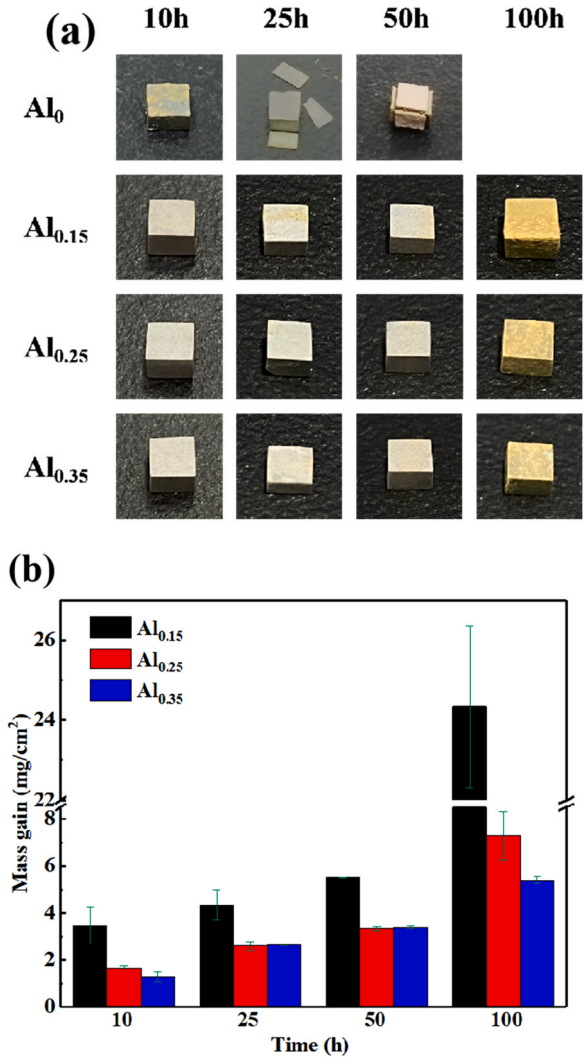


Fig. 6. Appearance (a) and the mass change (b) of samples after isothermal oxidation at 1000 ℃ for different times, respectively.

the sample surface. With the increase of oxidation time to 50 h, the three Al-bearing alloys still maintain good surface quality, which shows smooth and flat surface without obvious cracks even at the edges and corners. When the oxidation for 100 h, the three alloys show different appearances. Although the $Al_{0.15}$ alloy shows uneven surface, the $Al_{0.25}$ and $Al_{0.35}$ alloys still have good surface quality. It can be clearly seen from the oxidation weight gain of alloys shown in Fig. 6b that the weight of the three alloys after oxidation increases gradually with the increase of oxidation time. Among them, $Al_{0.15}$ alloy has more oxidation weight gain, and its mass increases by 24.3 mg/cm^2 after 100 h oxidation. With the increase of Al content, the trend of oxidation weight gain is slowed down significantly, and the mass increase of $Al_{0.25}$ and $Al_{0.35}$ after 100 h oxidation is only 7.3 mg/cm^2 and 5.4 mg/cm^2, respectively. In general, the addition of Al significantly improves the long-term oxidation resistance of the alloy. With the increase of Al content, the alloy has much better appearance and lower weight gain after oxidation. During the whole oxidation process, $Al_{0.25}$ and $Al_{0.35}$ alloys show similar oxidation resistance.

Fig. 7 shows the logarithmic curves of mass change of $Al_{0.15}$, $Al_{0.25}$ and $Al_{0.35}$ alloys after isothermal oxidation at 1000 ℃ for different times. According to the alloy oxidation growth theory [36], the mass change per unit area (ΔW/A) during isothermal oxidation over long periods of time can be explained as in Eq. (1), where n means the

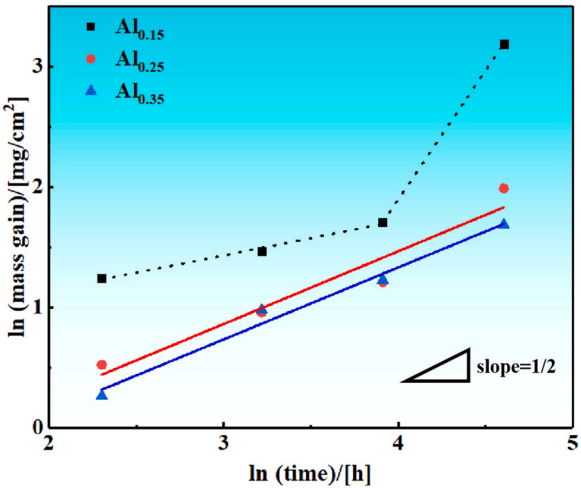


Fig. 7. Double-logarithmic plots of mass change against time of $Al_xCrTaTi$ alloys.

oxidation rate exponent, t is the time and k_n stand for the oxidation constant. Using the double-logarithmic plot of mass change versus time, the kinetic parameters, oxidation rate exponent n and oxidation constant k_n, can be evaluated according to Eq. (2).

$$(\Delta W/A)^n = k \cdot t \tag{1}$$

$$\ln(\Delta W/A) = \frac{1}{n} \ln k_n + \frac{1}{n} t \tag{2}$$

It is well known that when $0.5 < n < 1.5$, the oxide film growth of the alloy conforms to the linear law. When $1.5 < n < 2.5$, the oxide film growth of the alloy follows a parabolic law. When $2.5 < n < 3.5$, the oxide film growth of the alloy abides by the cubic exponential law. The alloys that generally conform to parabolic and exponential laws are considered to have much better oxidation resistance [37]. Table 2 summarizes the values of n, k_n and R^2 of the fitted alloy after isothermal oxidation at 1000 ℃ for 100 h. The closer R^2 is to 1, the higher the fitting degree is. It should be noticed that the n value of $Al_{0.15}$ alloy is different at different stage due to its different oxidation behaviors. It has a good oxidation resistance behavior before oxidation of 50 h with the exponential law. However, after oxidation of 50 h, the oxidation resistance decreases sharply with a linear rule. With the increase of Al content, $Al_{0.25}$ and $Al_{0.35}$ samples exhibit good antioxidant performance throughout the whole oxidation period. The n values of the two alloys are 1.65 and 1.67, respectively. This implies that the $Al_{0.25}$ and $Al_{0.35}$ alloys with the parabolic law produce the protective oxide film during the oxidation process and reduce the oxidation rate of the alloys.

In order to further analyze the oxidation mechanism of $Al_xCrTaTi$ RMEAs, the oxidation products on the surface after being oxidized at 1000 ℃ for 10 h were analyzed by XRD (see Fig. 8). The oxides of the original alloy Al_0 are mainly $CrTaO_4$, TiO_2 and Cr_2O_3. Besides, Al_2O_3 peaks are appeared in the Al-bearing alloys, indicating that the addition of Al causes the formation of Al_2O_3 in the oxidation products.

Table 2
Oxidation rate exponents (n) with regression coefficient (R^2) and n-dependent oxidation constants k, derived from a double-logarithmic plot using Eq. (2).

Alloy	n	R^2	Oxidation rate constant k
$Al_{0.15}$	(≤50 h)3.48*	0.99*	$k = 7.25 \cdot 10^0$[mg^2/cm^{4}h^2]*
	(≥50 h)− 0.17*	1*	$k = 3.11 \cdot 10^0$[mg^2/cm^{4}h^2]*
$Al_{0.25}$	1.65	0.94	$k = 2.08 \cdot 10^{-1}$[mg^2/cm^{4}h^2]
$Al_{0.35}$	1.67	0.98	$k = 1.69 \cdot 10^{-1}$[mg^2/cm^{4}h^2]

* Note, the oxidation of $Al_{0.15}$ alloy is reasonably followed by different oxidation laws in the two periods before and after oxidation for 50 h, and the segmenting fitting is adopted.

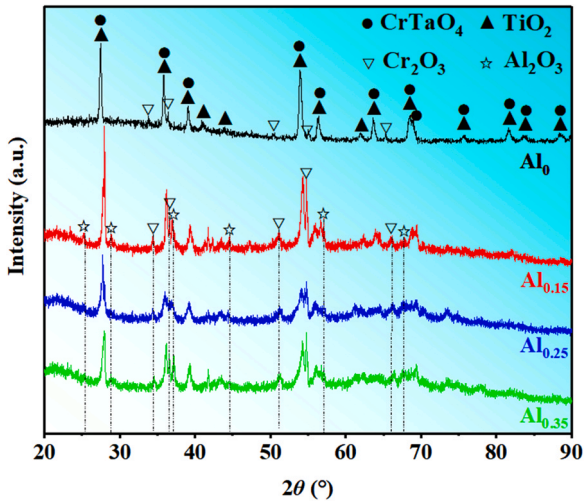


Fig. 8. XRD patterns of the oxidation products of $Al_xCrTaTi$ alloys after 10 h oxidation.

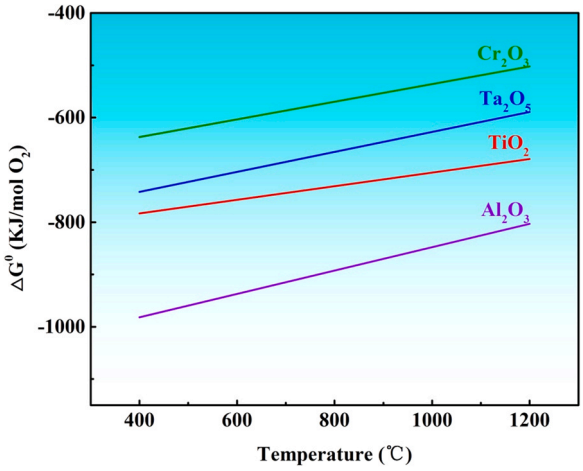


Fig. 10. Standard Gibbs free energies of the formation of several oxides at different temperatures.

Moreover, as the Al content decreases, the overall signal of the sample also decreases due to the thickness decrement of the oxide layer (given hereafter), resulting in a decrease in the Al_2O_3 peak intensity. To decode the effect of Al on the high-temperature oxidation behavior of CrTaTi alloy, Fig. 9 shows the backscatter electron images of the cross sections and element distribution of the four alloys after isothermal oxidation at 1000 ℃ for 10 h. The thickness of the oxide layer of Al-bearing alloys decreases dramatically compared with the Al-free alloy, where the oxide thickness of Al_0, $Al_{0.15}$, $Al_{0.25}$ and $Al_{0.35}$ alloys are 23.4 μm, 5.87 μm, 4.14 μm and 2.42 μm, respectively. In addition, the oxidation layer of Al_0 alloy is stratified, including the surface layer, intermediate layer and internal corrosion zone in the lower layer. The surface layer produces a convex area, and there is an obvious gap between the surface layer and the oxidation middle layer. The oxide layers of the Al-containing alloys show relatively dense and continuous without the delamination after the oxidation of 10 h. A deep inner oxidation zone can be observed under the oxide layer of all the experimental alloys. Combined with the elemental analysis, the oxide layers of the three Al-containing alloys are mixed oxides layers mainly containing Al_2O_3, Cr_2O_3 and TiO_2. However, the internal corrosion areas under the oxide layer are nitrides, mainly distributed in the interdendritic areas.

4. Discussion

The Al-free CrTaTi alloy preferentially formed a composite oxide layer mainly containing Cr_2O_3 and TiO_2 during the early stages of oxide growth with the surface of the oxide layer. With the increase of the oxidation time, a $CrTaO_4$-based oxide layer (the middle oxide layer, see Fig. 9a) is formed under the composite oxide layer. Fig. 10a shows the standard Gibbs free energy of formation of the oxidation products about several elements at different temperatures [38]. TiO_2 is preferentially generated because it has the most negative Gibbs free energy of formation of oxide at 1000 ℃ among the oxides of Cr, Ta and Ti. Although Ta_2O_5 has the second negative Gibbs free energy of formation of the oxide, Ta has a lower diffusion rate than Ti and Cr [39], and Ta has much larger atomic mass (181) than Ti (48) and Cr (52). In addition, the high entropy alloys possesses serious lattice distortion, and lead to a sluggish diffusion effect and need to overcome the continuous activation barriers. Therefore, the lowest effective diffusion efficiency and the slower kinetics makes it hard to

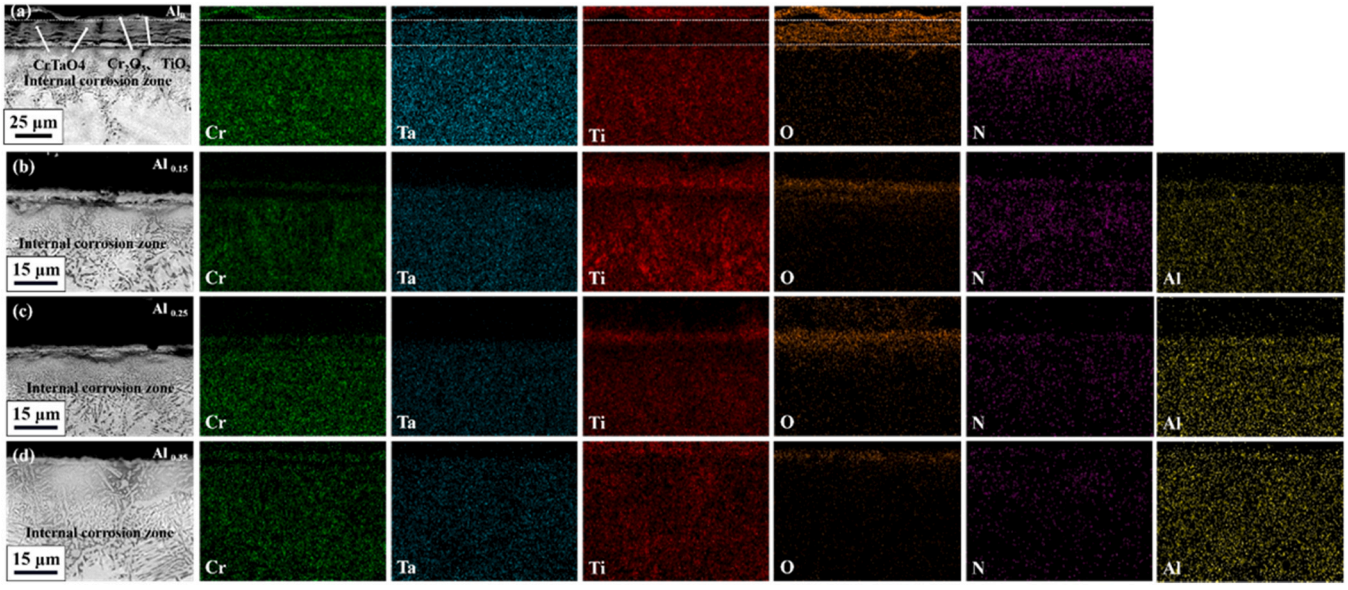


Fig. 9. Cross-sectional images and element distribution of oxide layer for $Al_xCrTaTi$ alloys exposed to the air at 1000 ℃ for 10 h.

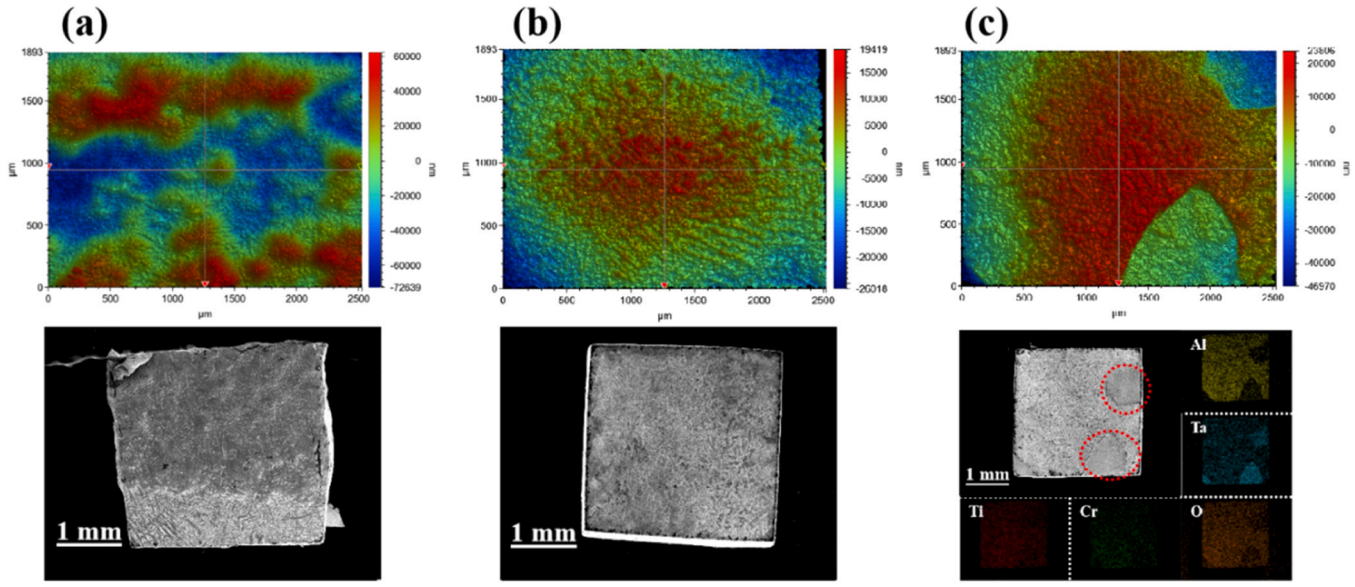


Fig. 11. The morphologies and surface roughness of three Al-bearing alloys after 100 h oxidation, and the elemental distributions on the surface of $Al_{0.35}$ alloy.

Table 3
PBR values of several oxides.

Oxide	Al_2O_3	TiO_2	$CrTaO_4$	Cr_2O_3
PBR	1.29	1.96	2.02	2.07

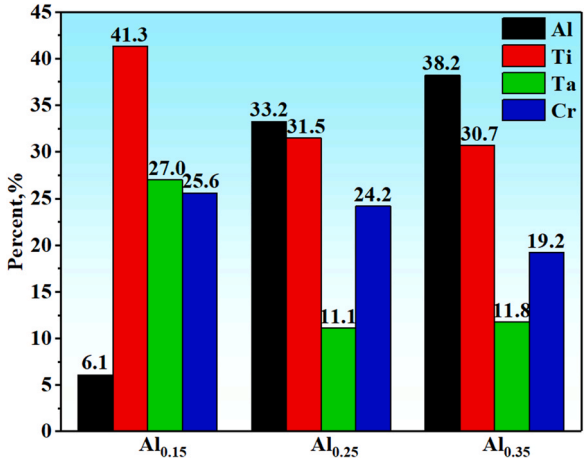


Fig. 12. The proportion of different elements on the alloy surfaces after removal of O for the Al-bearing RMEAs.

generate Ta oxides. Although the standard molar Gibbs free energy of formation of Cr_2O_3 is slightly higher than that of Ta_2O_5, Cr_2O_3 is easier to be preferentially generated due to a relatively large diffusion rate. Therefore, the composite oxide layer containing Cr_2O_3 and TiO_2 is preferentially generated on the surface oxide layer, which shows good oxidation resistance in the short term. But, the oxidation resistance of the composite oxide layer decreases with the increase of oxidation time due to TiO_2 with a large thermodynamic driving force and a strong affinity with oxygen. During the long-term oxidation, TiO_2 will grow faster than Cr_2O_3, which may cause the surface oxidation layer with crack due to the volume ratio imbalance of oxides, and eventually it will lose the antioxidant capacity. The $CrTaO_4$ is formed mainly following by the reaction of Cr_2O_3 and Ta_2O_5 at 1000 ℃ [40], which first requires an incubation period to generate Cr_2O_3 and Ta_2O_5. Because of Ta with a low diffusion rate, it delays the generation time of Ta_2O_5, resulting $CrTaO_4$ lagged in the lower layer [39]. It is found that $CrTaO_4$ has a very low growth rate at 1000 ℃, and its growth rate constant is only 3.83×10^{-17} [41], even lower than that of protective Cr_2O_3 (8.08×10^{-17} [42]). In addition, $CrTaO_4$ can effectively prevent the oxidation of RMEAs by inhibiting the outward diffusion of cations [41], showing high antioxidant activity. However, the whole oxidation process of CrTaTi alloy is dominated by the inward diffusion of oxygen. With the increase of oxidation time, the composite oxide layer on the surface loses its protective property due to the rapid growth of TiO_2, which leads to the cracking of the oxide layer. Although $CrTaO_4$ is an oxidation product with good oxidation resistance, the oxidation process of CrTaTi alloy is dominated by the inward diffusion of oxygen [41], making the Al-free alloy with poor long-term antioxidant properties.

The high-temperature long-term oxidation resistance of Al-containing alloys is significantly improved, which is mainly attributed to the presence of Al_2O_3 in the oxide layer of the alloys. Fig. 10shows Al_2O_3 with the lowest Gibbs free energy of formation than the other three oxides, implying that Al_2O_3 will be generated firstly during oxidation. Further, the preferentially formed Al_2O_3 as the nucleation site promotes the formation of a continuous protective Cr_2O_3 oxide layer [43], which is confirmed by the enhancement of Cr_2O_3 peak within the oxide layer of Al-containing alloys shown in Fig. 8. The valence electron number (VEC) of several alloying elements is 3 (Al), 4 (Ti), 5 (Ta) and 6 (Cr), respectively. It is well known that for metallic elements, as the number of valence electrons increases, their tendency to be oxidized decreases because the higher energy is required for electrons to leave its parent atom [44]. This is similar to the electronegativity of metals. Therefore, Al with lower VEC is preferential oxidation, which is consistent with some previous reports [45,46]. Therefore, the presence of Al_2O_3 in the oxide layers of the three Al-bearing alloys greatly improves the oxidation resistance. In addition, the delamination of the oxide layer for the Al-bearing alloys after 10 h oxidation was not clearly observed due to the formation of the relatively continuous and dense Al_2O_3-containing composite oxide layer. However, a layer of nitriding zone can still be observed on the matrix of the three alloys because the generated Al_2O_3 is not continuous and cannot completely prevent the inward diffusion of oxygen and nitrogen.

In order to further explore the oxidation mechanism of Al-bearing RMEAs, the surface roughness of the oxide layer after 100 h oxidation was analyzed. The appearance morphologies of the oxide layer of Al-containing alloys are very different. Fig. 11a exhibits the

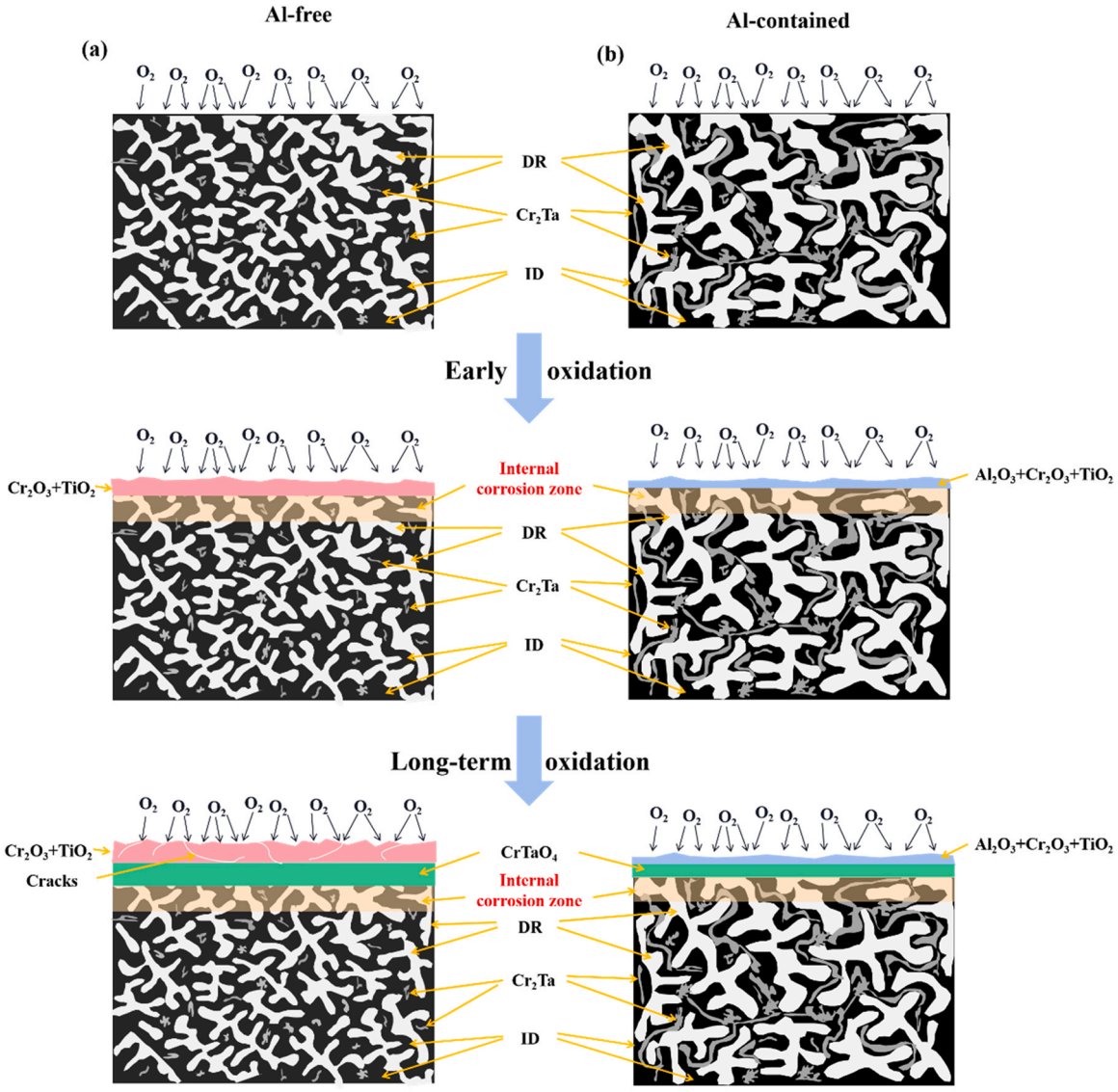


Fig. 13. Schematic diagram of oxidation mechanism of Al-free and Al-bearing RMEAs.

uneven surface of $Al_{0.15}$ alloy, indicating that a large number of oxidation products are generated on the raised part. With the increase of Al, the surface height of $Al_{0.25}$ alloy has little difference and the surface becomes smooth (see Fig. 11b). With the further increase of Al content, although the overall surface of $Al_{0.35}$ alloy also shows a good flatness, the upper right and lower right corner of the sample obviously shows the peeling off of the oxidation layer (red circle part), exposing more yellow oxidation products in the inner layer (see Fig. 11c). This peeling phenomenon of the oxidation layer is caused by the use of tweezers for observation samples. The mismatch of the thermal expansion coefficient between the oxide layer and matrix would lead to the increasing of stress at the interface during cooling. The density oxide layer of $Al_{0.35}$ gives rise to the concentration and unreleased of stress at the interface, resulting the oxide layer has weaker adhesion and can be peeled under a slight force. The peeling behavior of the oxide layer in $Al_{0.35}$ further indicates the weaker adhesion between the oxide layer and matrix than the other Al-bearing alloys. However, this situation will not occur in $Al_{0.15}$ and $Al_{0.25}$ samples, indicating that the very poor adhesion between the oxide layer and the matrix of $Al_{0.35}$ alloy. Furthermore, the distribution of elements from the product layer to the spalling position are changed from Al-rich and Ti-rich to Ta-rich, and Cr is evenly distributed on the entire sample surface. This indicates that the oxidation products on the sample surface contain Al_2O_3, TiO_2 and Cr_2O_3, and the exposed oxidation layer is mainly $CrTaO_4$. This phenomenon shows that the upper layer of composite oxide layer is Al_2O_3, TiO_2 and Cr_2O_3, and the lower layer is $CrTaO_4$, which is similar to the original alloy Al_0, indicating that Al does delay the formation of $CrTaO_4$ [47].

Generally, the uneven surface morphology of oxide layer is related to the growth rate and PBR (Pilling-bedworth ratio, PBR) value of oxidation products [48]. The oxidation products of Al-bearing alloys are mainly TiO_2, Al_2O_3, Cr_2O_3 and $CrTaO_4$. The oxidation growth constant of Al_2O_3, Cr_2O_3 and $CrTaO_4$ is very low, which means that the growth rate is very slow. TiO_2 has a faster growth rate due to its greater affinity with oxygen and relatively larger thermodynamic driving force [38]. The PBR values of several oxidation products are listed in Table 3. The PBR value of TiO_2 is at a high level among several oxidation products. Therefore, with the increase of oxidation time, TiO_2 with a faster growth rate grows faster in the oxidation layer, and its large PBR value leads to the uneven oxidation layer of $Al_{0.15}$ alloy. Because Al_2O_3 has a lower standard Gibbs free energy of formation, Al_2O_3 will be preferentially generated under the same conditions. With the increase of Al content, the generation of enough

Al_2O_3 on the alloy surface will inhibit the generation of TiO_2 to a certain extent, resulting the surface morphologies of $Al_{0.25}$ and $Al_{0.35}$ alloys are relatively smooth and flat.

Fig. 12 shows the element content on the oxide layer surface of Al-bearing alloys after oxidation for 100 h with removal of O. The Ti content of oxidation products on the alloy surface is gradually decreased with the increase of Al, which also indicates that the excessive TiO_2 is indeed one of the reasons for the uneven surface of $Al_{0.15}$ alloy. At the same time, the addition of Al also inhibits the formation of TiO_2 to a certain extent. The Al content of the oxidation products on the alloy surface is reasonably gradually increased with the addition of Al. Although it can improve the oxidation resistance of the alloy, the overall oxidation resistance of the alloys is not only related to the formation of a dense continuous and protective oxide layer, but also the adhesion of the oxide layer on the substrate. Within the oxidation products of AlxCrTaTi alloy system, Al_2O_3 has the lowest PBR value. When the Al_2O_3 content of the oxidized surface layer increases, the overall PBR value of the surface layer will decrease, resulting it is easier to imbalance with the volume ratio of the intermediate layer mainly containing $CrTaO_4$ with a large PBR value. With the increase of oxidation time, the larger difference of PBR values of oxides leads to the interfacial stress concentration between the top layer and the bottom layer of the oxide layers, which makes them easy to be peeled off and deteriorates the oxidation resistance of the alloy [48]. In addition, the large lattice mismatch between α-Al_2O_3 with a hexagonal structure and TiO_2 with a rutile structure can also cause a decrease in the adhesion of the surface oxide layer [49]. In general, $Al_{0.25}$ alloy has both good room-temperature mechanical properties and high-temperature oxidation resistance in the Al_xCrTaTi RMEAs.

Fig. 13 presents a schematic diagram of the oxidation mechanism of Al-free and Al-bearing alloys. For the original CrTaTi alloy as shown in Fig. 13a, the composite oxide layer containing Cr_2O_3 and TiO_2 is preferentially formed at the early stage of oxidation, and a layer of nitrogen-rich inner oxidation zone can be clearly observed below the oxide layer. With the increase of oxidation time, the oxide layer mainly $CrTaO_4$ grows below the composite oxide layer. Due to the rapid growth of TiO_2, the cracks are produced by the stress concentration of surface oxide layer, where nitrogen and oxygen can continue to enter. Although $CrTaO_4$ has good antioxidant capacity, it cannot effectively prevent the internal diffusion of nitrogen and oxygen, resulting a relatively poor long-term oxidation resistance of the Al-free CrTaTi alloy. For the Al-bearing alloys as shown in Fig. 13b, the composite oxide layers containing Al_2O_3, Cr_2O_3 and TiO_2 are preferentially generated on the surface during the early oxidation stage. The considerable amount of Al_2O_3 with the thin oxide layer can effectively prevent the oxidation. However, nitrogen and oxygen can still enter the alloys due to the formation of discontinuous Al_2O_3, resulting the formation of nitriding zone under the oxide layer. As the oxidation time increases, $CrTaO_4$ is also generated under the surface oxidation layer, which can well prevent the outward diffusion of metal cations. In addition, the dense and continuous composite oxide layer containing Al_2O_3 on the surface can prevent anions such as nitrogen and oxygen from spreading inward to a certain extent. The synergistic effect of the two oxide layers makes the alloy show good oxidation resistance even after long-term oxidation.

5. Conclusions

In this work, Al_xCrTaTi refractory medium-entropy alloys with different Al contents were developed. The effects of Al on the microstructure, room-temperature mechanical properties and high-temperature oxidation behavior were systematically studied. The main conclusions are as follows:

(1) The addition of Al promotes the formation of Cr_2Ta Laves phase in Al_xCrTaTi refractory medium-entropy alloys. With the increase of Al, the Cr_2Ta phase gradually increases. In addition, the shape of Cr_2Ta phase changes from the rice-grained distribution to gradually growing and lengthening, and finally connecting to each other in the interdendritic region with a spider network.
(2) The addition of Al makes the severe lattice distortion and promotes the formation of Cr_2Ta phase, thus improving the microhardness and yield strength of the alloy. Due to the introduction of low-density Al, the Al_xCrTaTi refractory high entropy alloys also exhibit high specific strength.
(3) The addition of Al greatly improves the high-temperature oxidation resistance of the alloys. After long-term oxidation at 1000 ℃ for 100 h, the oxidation weight gain of $Al_{0.25}$ and $Al_{0.35}$ alloys are all less than 10 mg/cm^2, and the growth of the oxide film conforms to the parabolic law. However, the alloy with excessive addition of Al will reduce the adhesion of the oxide layer surface, resulting the peeling-off of oxide layer under the action of external force and the decline of the oxidation resistance.
(4) Among the Al-bearing alloys, $Al_{0.25}$ alloy shows the best comprehensive properties, where the room-temperature microhardness, yield strength and fracture strain are 679 HV, 1669 MPa and 12.8%, respectively. The oxidation weight gain is only 7.3 mg/cm^2 after oxidation at 1000 ℃ for 100 h.

CRediT authorship contribution statement

Hongju Zhang: Validation, Investigation, Data Curation, Methodology, Visualization, Writing – review & editing. **Yuhang Du**: Data curation, Writing – original draft preparation, Visualization, Investigation; **Limin Lai:** Validation, Investigation, Writing – review & editing. **Ning Guo:** Investigation, Methodology; **Ning Li:** Methodology, Conceptualization; **Shengfeng Guo:** Supervision, Conceptualization, Methodology, Writing – review & editing, Funding acquisition.

Data Availability

Data will be made available on request.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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